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CLASSIFICATION OF ELECTRONIC ATLASES BY THE METHOD OF PAGE PRESENTATION

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Electronic atlases continue to serve as one of the primary tools for knowledge dissemination and the presentation of research findings in earth sciences and geography. At the beginning of any electronic atlas conceptual design phase, authors need to determine the number of thematic sections and, as a result, the number of pages and the way they are presented. The latter aspect has received insufficient attention, and the available options for implementation have not been formalized. To develop a classification of electronic atlases by the method of page presentation, a sample of 112 electronic atlases was compiled. The main criteria for the selection of electronic atlases were: 1) the presence of the word 'atlas' in the title; 2) functioning as of August 2024; and 3) the presence of geographic or spatially referenced content in any form (texts, maps, charts, multimedia, etc.). As a result, six classes of electronic atlases were identified based on the method of page presentation and grouped into three types: single-page, pseudo-single-page, and multi-page. The purpose of further research is to develop guidelines on 1) when to choose one of the three types of electronic atlases and 2) how to design each type.

Keywords: digital atlas, atlas cartography, geosciences, classification of atlases, single-page atlas, multi-page atlas.

Introduction

An important component of scientific activity is the presentation of research findings to colleagues and the general public [6]. In Earth Sciences this function is usually provided by geoportals (e.g., Denmark's Geology Portal) or map viewers. Such applications allow users to efficiently download and visualize data sets as analytical maps. However, the combination of datasets is left to the user's choice, and additional explanations are limited to metadata and/or short formal texts. Electronic atlases (EAs), rather than relying solely on ready-made multivariate and synthetic maps, may combine various multimedia presentations emphasizing the connections between content elements in a narrative way. At the same time, the meta-concepts of EAs are not limited to collections of maps/modules or storytelling atlases [1]. If maps are not the primary content, the meta-concept of EA as an organizational mechanism may be applied, using a single map as an interface for accessing spatially referenced multimedia representations. Developing the most suitable concept of concrete EA requires a general understanding of modern types of EA.

Classifications of EAs are covered in detail in the works of Ormeling [7], Borchert [2], Schneider [10], and Hurni [4]. Among the classification features specific to EAs are the following: platform (PC or mobile devices); functionality/interactivity [7]; spatial dimension of cartographic representation (2D, 2.5D, and 3D) [10]; type of data storage/transfer (online, hybrid or standalone application). Nowadays, this list needs to be supplemented with a number of features related to system architecture, interface, information architecture (IA) and navigation, content representation, functionality/interactivity, and EA as a whole. In this publication, we will focus on just one fundamental characteristic of EAs: the method of page presentation.

In the 1990s and early 2000s, the most common were EAs, which, like paper atlases, consisted of many separate pages. However, in the period 2007–2016, the number of EAs whose content was opened on a single page or single screen increased significantly. Currently, the number of single-page EAs has slightly decreased, but is still substantial. Although EAs remain a combination of different pages or windows with various forms of information representation, which can be conditionally considered pages [11], Web 2.0 technologies have enabled dynamic (asynchronous) updates of a web page without reloading, thereby creating the effect of a single-page application. At the same time, there appeared the problem of distinguishing between a single-page EA with a flexible concept of map information provision [9] and web maps and other cartographic information systems that can also consist of multiple thematic layers or maps.

The classification of EAs by the method of page presentation is not an academic exercise. The choice of the type of EA by the method of page presentation should be made at the very beginning of the conceptual design, as it determines the details of creating at least the IA, navigation, and interface of the EA. For instance, for a multi-page EA, it is necessary to develop a top-level IA (the EA as a website) and an IA of the map or thematic section. Determining the number of pages with a unique interface (screens) implies the development of layouts for such pages. In other words, the design of a multi-page and a single-page EA is noticeably different and likely requires different design guidelines. To confirm this statement, it is necessary to first identify and analyze the existing types of EAs by the method of page presentation.

Materials and Methods

A primary methodological challenge was establishing criteria for a sample of EAs. EAs are

created in various disciplines and may have different types of content elements and different functions. However, the essential features of EA as a concept have not yet been defined at an interdisciplinary level. The prevailing opinion among cartographers is that atlases should belong to the field of Earth Sciences, consist mainly of maps, and not be encyclopedias [13]. For EAs, these requirements are complemented by the presence of multiple interactive maps across multiple pages and the focus on a non-expert audience [1]. However, more authoritative Swiss cartographers insist on a single map interface and intrinsic storytelling for EAs [12].

The prepared sample of EAs (see the 'Materials and research methods' section of the article [5]) is based on only three criteria: 1) the presence of the word 'atlas' in the title of EAs (or calling it an 'atlas' by the authors in the description or other publications); 2) the availability of the EA for viewing or downloading as of August 2024; 3) the presence of geographic or spatially referenced content in any form (texts, maps, charts, multimedia, etc.). This set of criteria enables us to take into account a wide range of possible interpretations of EA in Earth Sciences, geography, and cartography. The sample comprises 112 EAs, identified through an analysis of scientific publications, primarily in cartography, and Google searches using queries such as 'Atlas of', 'Digital atlas of', 'Online atlas of', 'Climate atlas of', and similar terms.

Results and discussion

Three types of EAs were identified according to the method of page presentation: single-page, pseudo-single-page, and multi-page. The relationship between these types is represented as a continuum (Fig. 1), where single-page and multi-page EAs are located at the extremes of the range, and pseudo-single-page EAs encompass classes within the continuum.

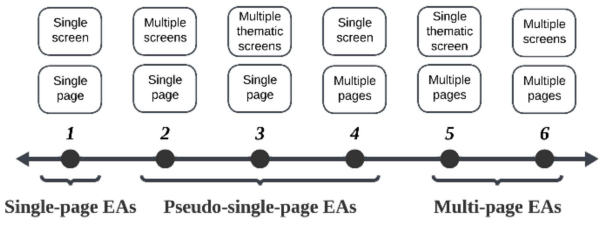


Fig. 1. Classification scheme of electronic atlases by the method of page presentation

We have categorized EA pages into the following types: home page, navigation pages, reference pages, content pages, overview (introductory) pages, and map pages. The key function of the EA home page is to describe the purpose and content of the EA, to interest the user, and to prepare him/her for working with the EA. Navigation pages do not contain thematic content and are used only as an access mechanism to EA elements. Often, these are pages with a detached navigation model [8]. Reference pages (e.g., About, Help, News, etc.) are not part of the thematic content. Their purpose is to provide information about EA and to help users master EA.

Content pages are all pages of the EA that contain thematic information (articles, maps, multimedia, etc.). In turn, content pages may include overview pages and map pages. Overview pages summarize the main points of the EA chapters with explanatory texts and navigation elements or using a dashboard-style approach. Possible types of overview pages ('entry pages') are discussed by Wang et al. [14], albeit in the slightly different context of visualization atlases. Map pages may include standalone map interfaces or pages with embedded maps [3]. It should be noted that information displayed in additional windows or sidebars of the map interface was not considered pages, as in Schulz's [11] interpretation. The new page implies a complete content replacement and/or reloading. For instance, the Help section will be considered a reference page only if it is opened on a separate page instead of the map page.

Single-page EAs refer to applications that display all their content on a single page in the same interface (only one screen). Single-page EAs include EAs with a layout in which the map occupies the entire screen space (Atlas of Prague, Statistical Atlas (Eurostat)), as well as with a fragmented layout consisting of multiple views: Atlas of Radioactive Contamination of Ukraine, Copernicus Interactive Climate Atlas. The concept of map information provision can be both restrictive and flexible [9]. The key is that the page refreshes without reloading.

At the beginning of the range of pseudo-single-page EAs are EAs whose only thematic or map section functions as a single-page application, but the EA already contains a separate home page and/or other non-thematic pages: Atlas of the Historical Geography of the United States, Social Capital Atlas. Such EAs require the design of multiple screens.

It is necessary to focus on EAs that consist of several modes/views. For example, in the OECD Regions and Cities Statistical Atlas, users can simply switch views or representation modes in the same interface (it is a single-page EA). In the Digital Atlas of Belgium, only the mode of functionality changes. Instead, the Atlas do Censo Demográfico 2010 and the IPCC WGI Interactive Atlas offer two different modes in terms of functionality, representation, navigation, and partly interface (it remains coherent, at least in terms of visual design). However, these EAs function as single-page applications, and switching modes occur without reloading, using the tabbed principle. These pseudo-single-page EAs already consist of multiple thematic screens.

The last class of pseudo-single-page EAs is composed of EAs that consist of multiple maps or pages that open on a single page of a unified

interface (single screen – multiple pages), but the page refresh is not asynchronous (the page reloads) and the state of the maps view is not saved during the session (e.g., Forest Atlas of the United States, Lake Huron Treaty Atlas). A unified interface means that both the layout (the manner in which the elements of each page are arranged and the character of the relationships between these elements) and visual design of the interface are not only coherent but almost identical (an interface template or pattern is applied). More common are EAs whose (homogeneous) thematic content is presented on separate pages in a unified interface ('single thematic screen'), but the EA also contains non-thematic pages: Alters Atlas, ÖROK-Atlas. EAs with such characteristics can already be considered multi-page.

Explicit multi-page EAs are EAs consisting of multiple thematic sections (individual pages or groups of pages) with different forms of information presentation (Climate Atlas of Canada), including different screens of map pages (Climate Impact Atlas). This type also includes EAs that are encyclopedias, i.e., consisting of only an article collection (Atlas Nacional de Espana) or articles together with a map interface section (Minnesota Breeding Bird Atlas). For EAs with an encyclopedia meta-concept, it is important to ensure that there are two-way links between the map and articles (the possibility of article-map navigation and vice versa), as implemented in the Minnesota Breeding Bird Atlas.

Lastly, we emphasize the distinction between EAs as multi-page websites and websites that only provide information about EAs but are not EAs themselves. A distinctive feature of the EA as a website is the presentation of the EA's thematic content directly on the website's pages. In contrast, the EA's website or portal contains comprehensive information about the EA and links to all versions of the EA without acting as a self-contained EA (e.g., Nationalatlas.de).

Conclusions

The analysis of the sample of 112 EAs revealed six possible classes of EAs based on the method of page presentation. These six classes were subsequently grouped into three primary types: single-page, pseudo-single-page, and multi-page. Although the results are based on empirical evidence, this does not mean that all six classes of EAs are optimal. First of all, it is necessary to determine the feasibility of creating pseudo-single-page EAs, in particular, of the second and fourth classes. For instance, in which cases is the addition of a home page screen an advantage, and whether the pseudo-single-page nature of the EA is not due to technical factors alone? A general issue is the development of guidelines that would indicate when and why it is desirable to choose single-page EAs or, on the contrary, multi-page EAs. Finally, it is extremely important to hear the users' opinion and determine which EAs are preferred in terms of the method of page presentation. A valid answer to this question can only be obtained based on the results of complex user experience studies of EAs.

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КЛАСИФІКАЦІЯ ЕЛЕКТРОННИХ АТЛАСІВ ЗА СПОСОБОМ ПРЕДСТАВЛЕННЯ СТОРІНОК

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Електронні атласи продовжують слугувати одним з основних засобів поширення знань та представлення результатів досліджень у галузі наук про Землю та географії. На початку проєктування будь-якого електронного атласу розробникам необхідно визначити кількість тематичних секцій та, як наслідок, кількість сторінок та спосіб їх представлення. Останньому аспекту досі приділялося недостатньо уваги, а наявні варіанти реалізації не зафіксовані. Для розробки класифікації електронних атласів за способом представлення сторінок було сформовано вибірку зі 112 електронних атласів. Основними критеріями відбору електронних атласів були: 1) наявність у заголовку слова «атлас»; 2) функціонування станом на серпень 2024 р.; 3) наявність географічного або просторово прив'язаного контенту у будь-якій формі (тексти, таблиці, графіки, карти тощо). У результаті було виявлено шість класів електронних атласів за способом представлення сторінок, які були об'єднані у три типи: односторінкові, псевдоодносторінкові та багатосторінкові. Метою подальших досліджень є вироблення рекомендацій щодо 1) вибору одного з трьох типів електронних атласів та 2) створення кожного типу атласів.

Ключові слова: електронний атлас, атласна картографія, науки про Землю, класифікація атласів, односторінковий атлас, багатосторінковий атлас.